

Distribution-Aligned Synthetic Text Generation via Tail-Aware Enhancement

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Abstract

Recent advances in generative AI have popularized synthetic content for training, offering a practical alternative to costly data curation while addressing privacy concerns. However, accumulating evidence shows that the indiscriminate reuse of synthetic data can induce *model collapse*—a degenerative process that contracts the learned distribution and erodes rare features. For instance, when models are iteratively trained on their own synthetic outputs, the upper tail of the perplexity distribution substantially compresses, with high-percentile values dropping by nearly half—a clear indicator of severe diversity loss.

To counter this, we introduce DASGEN, a **D**istribution-**A**ligned **S**ynthetic **T**ext **G**eneration framework via tail-aware enhancement. Our method first identifies underrepresented regions via embedding-space mining, then steers a frozen, hosted LLM using semantically-structured prompts and a discriminative diversity objective to enrich tail features. This training-free approach enables direct deployment in existing data pipelines. Extensive evaluations on Yelp and ICLR’25 review benchmarks show that DASGEN significantly outperforms competitive baselines, achieving tail coverage (98.54% on Yelp; 92.00% on ICLR’25) along with improved downstream accuracy. Overall, DASGEN provides a practical path to synthesizing distribution-aligned text by explicitly enhancing tail regions, producing synthetic corpora with enhanced coverage and diversity for more reliable long-tailed applications.

CCS Concepts

• **Security and privacy** → **Privacy protections**.

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Keywords

synthetic data; long-tail coverage; LLMs

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1 Introduction

With the rapid advancement of generative AI, particularly large language models (LLMs), synthetic data has emerged as a pivotal resource for model training and adaptation [19, 26, 44]. Beyond serving as a supplementary corpora for pre-training [1, 30], synthetic data are increasingly viewed as a scalable and privacy-preserving alternative to real-world data, especially in domains where data acquisition is costly, sensitive, or constrained by regulation [15, 22]. For instance, in autonomous driving, collecting sufficient training data for rare scenarios like extreme weather conditions or emergency maneuvers incurs prohibitive costs and safety risks—synthetic data generation offers a viable solution by creating diverse driving scenarios programmatically [25, 40]. Similarly, in healthcare, where patient data privacy is paramount and sharing real medical records faces strict regulatory barriers, synthetic clinical notes and medical images can preserve statistical fidelity while removing identifiability concerns [41, 47]. These advantages make synthetic data a practical resource for fine-tuning foundational models, and improving their performance across tasks such as complex reasoning [37, 42], domain-specific understanding [51], spatial cognition [21], and question answering [31, 51].

However, recent studies [6, 7, 12, 39] highlight a critical risk in this paradigm: model collapse, a degenerative process in which models iteratively trained on their own synthetic outputs exhibit progressively degraded generalization. As illustrated in Figure 1, human-written text exhibits a broad perplexity distribution, while

iteratively generated samples (Gen_1 $\rightarrow$ Gen_5) become progressively sharper, with high-percentile mass markedly reduced. This tail compression signifies a critical distributional narrowing: frequent patterns are over-represented while rare, information-rich semantics are systematically under-covered—a mismatch that directly degrades model performance on long-tailed tasks. These observations compel the development of distribution-aligned synthetic text generation frameworks, which explicitly target under-represented tail semantics to preserve diversity and reliability.

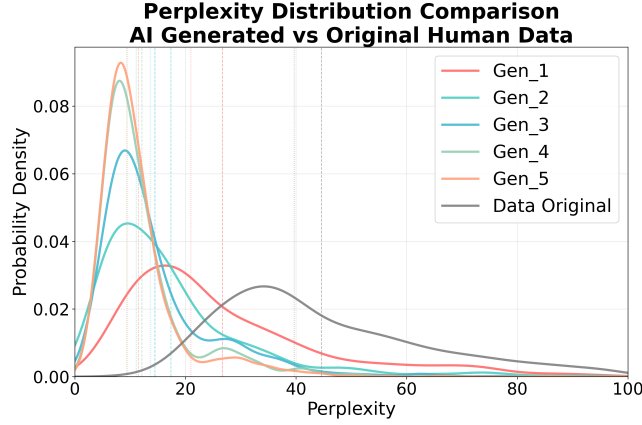


Figure 1: Perplexity Distribution Comparison between AI-Generated and Human-Written Data. Following the setup of [39], Gen_i is trained using synthetic data generated by Gen_{i-1}.

While prior works [14, 39] have investigated the impact of synthetic data on model robustness and performance, practical solutions to counteract distributional narrowing at the data generation phase remain underexplored. Existing approaches, such as heuristic rephrasing [52] or naive prompt engineering [18, 27, 36, 46], primarily operate on the surface-level variation of text. They lack a systematic, representation-level mechanism to guarantee sufficient coverage of semantically sparse regions—precisely the long-tail features essential for maintaining generalization and fairness in downstream models. This limitation underscores the need for a generation paradigm that moves beyond superficial diversity to explicitly model and enhance the underrepresented semantics in the embedding space.

To this end, we introduce DASGEN, a tail-aware synthetic data generation framework designed to explicitly align synthetic distributions with real-world data by enhancing underrepresented semantics. Our approach operates in two coordinated stages: it first performs lightweight statistical analysis on the semantic embeddings to systematically identify sparse, underrepresented regions that correspond to long-tail concepts. Then, we steer a frozen, hosted LLM using structured semantic prompts and a discriminative diversity objective, explicitly guiding the generator to produce text that incorporates these identified tail semantics. By integrating this distributional awareness directly into the data synthesis

pipeline, DASGEN promotes a more balanced and diverse data coverage, mitigating model collapse and enabling the construction of robust models even under high synthetic-data regimes.

Contribution. Our main contributions are summarized as follows:

- We propose an efficient method to identify semantically sparse dimensions in the embedding space. This allows us to systematically detect tail regions that are often underrepresented in synthetic data.
- We design a training-free, model-agnostic generation mechanism that explicitly enhances long-tail coverage through tail-aware prompting, ensuring distributional alignment without fine-tuning generator weights—an advantage for deployment in privacy-sensitive and computation-constrained settings.
- Through extensive experiments on two benchmark datasets, we demonstrate that our framework achieves tail coverage reaching 98.54% on Yelp and 92.00% on ICLR’25, and yields synthetic corpora with markedly improved diversity, leading to superior downstream accuracy compared to state-of-the-art baselines.

2 More Related Work

2.1 Synthetic text generation

Synthetic text generation has evolved through several distinct paradigms. Early rule-based and template-driven systems provided high controllability and label consistency but suffered from poor scalability and limited linguistic diversity [13, 17]. Neural generative models, such as sequence-to-sequence architectures and attribute-conditioned frameworks, improved fluency but were restricted by the coverage of their training corpora [48]. The advent of large-scale pre-trained language models (PLMs) has established a dominant paradigm for synthetic text generation, enabling high-quality, scalable sampling that is widely used in data augmentation and fine-tuning [3, 45]. However, a critical limitation of this paradigm is its focus on surface-level fluency and diversity, often at the expense of deeper distributional fidelity. Without explicit mechanisms to ensure alignment, these powerful generators can inadvertently amplify biases present in their training data or prompts, particularly leading to the erosion of information in long-tail regions of the data distribution.

2.2 Mitigating Distributional Drift

Long-tail distributions are pervasive in real-world data, where a few head classes dominate while many rare but important categories remain underrepresented [2, 50]. Standard models not only inherit this imbalance but also tend to amplify it during training, further weakening the representation of tail features [16, 24]. When such models are later used to generate synthetic content, this bias is compounded, leading to outputs that overfit common patterns while neglecting long-tail semantics. A significant body of research addresses distributional drift in models trained on synthetic data. A widely adopted strategy is to continually mix real data during iterative training to anchor the model to the original distribution [6, 7, 12, 39]. While effective, this approach’s dependency on ongoing access to real data limits its use in privacy-sensitive or data-scarce scenarios. Alternative methods employ post-hoc filtering and weighting to curate synthetic datasets by removing low-quality samples or promoting diversity [5, 52]. However, these strategies

typically rely on heuristic utility functions that are not designed to explicitly preserve the statistical structure of the long tail. Although model-level interventions like distribution-aware fine-tuning [33] can help, they often require costly retraining and do not provide a holistic solution for distributional alignment. Critically, these mitigation strategies largely operate without an explicit objective to enhance underrepresented tail semantics.

2.3 Tail-Aware Enhancement in Data Generation

The challenge of long-tail learning is well-established, with standard models notoriously amplifying the inherent imbalance and underperforming on rare categories [24, 29]. In computer vision, a rich set of tailored solutions has emerged, which can be categorized as: (i) generative augmentation for tail classes [43]; (ii) representational compensation using external semantics [34]; and (iii) distribution-aware regularization [33]. These methods conclusively demonstrate that explicitly reinforcing tail regions is essential for robustness. However, their direct application to text generation is hindered by the discrete nature of language and the imperative for global semantic coherence, creating a gap for text-specific tail-aware enhancement techniques.

In summary, existing work provides robust methods for generating fluent text and for addressing long-tail imbalances in continuous domains. However, a critical gap remains at their intersection: current synthetic text pipelines lack principled and training-free mechanisms to ensure distributional alignment by explicitly preserving long-tail features. Our work, DASGEN, is designed to bridge this gap. We introduce a framework that performs embedding-space mining to identify underrepresented regions and employs tail-aware generation steering to explicitly enhance their coverage, ensuring the synthetic corpora are both high-quality and faithful to the long-tail characteristics of real-world data.

3 Preliminaries

3.1 Differential Privacy

Differential Privacy (DP) provides a rigorous mathematical framework for quantifying the privacy guarantees of algorithms by bounding the difference in output distributions over neighboring datasets.

Definition 3.1 (Neighboring Datasets). Two datasets $\mathcal{D} = (z_1, \dots, z_n)$ and $\mathcal{D}' = (z'_1, \dots, z'_n)$ over the same domain $\mathcal{Z}$ are *neighboring* (denoted $\mathcal{D} \sim \mathcal{D}'$) if they differ in exactly one record. That is, there exists an index i^* such that $z_{i^*} \neq z'_{i^*}$, and $z_i = z'_i$ for all $i \neq i^*$ [10, 11].

THEOREM 3.2 (DIFFERENTIAL PRIVACY). [9] Let $\mathcal{D}, \mathcal{D}'$ be neighboring datasets. Given parameters $\epsilon > 0$ and $\delta \in [0, 1]$, a randomized algorithm $\mathcal{M} : \mathcal{X} \rightarrow \mathcal{Y}$ satisfies (ϵ, δ) -differential privacy if for all measurable subsets $S \subseteq \mathcal{Y}$,

$$\Pr[\mathcal{M}(\mathcal{D}) \in S] \leq e^\epsilon \Pr[\mathcal{M}(\mathcal{D}') \in S] + \delta.$$

When $\delta = 0$, $\mathcal{M}$ satisfies ϵ -differential privacy.

THEOREM 3.3 (SEQUENTIAL COMPOSITION). [9] Let $\mathcal{M}_1$ and $\mathcal{M}_2$ be mechanisms satisfying ϵ_1 -DP and ϵ_2 -DP, respectively. Then their composition $\mathcal{M}(\mathcal{D}) = (\mathcal{M}_1(\mathcal{D}), \mathcal{M}_2(\mathcal{D}))$ satisfies $(\epsilon_1 + \epsilon_2)$ -DP. That

is, DP guarantees are preserved under any post-processing of the mechanism's output.

3.2 Notation and Problem Definition

Let $\mathcal{D}_p = \{x_i\}_{i=1}^N$ denote the private text dataset, where each sample x_i is encoded into embedding vectors $\mathbf{e}_i = f_\theta(x_i)$ using a pre-trained encoder f_θ . Our goal is to generate a synthetic dataset $\mathcal{D}_s = \{\tilde{x}_j\}_{j=1}^M$ that achieves both *distributional alignment* with $\mathcal{D}_p$ and *privacy preservation* under the (ϵ, δ) -differential privacy framework.

3.3 Threat Model

We adopt a practical threat model where the text generation API provider operates under the *honest-but-curious* adversary assumption. While the provider faithfully executes requested services, it may attempt to infer sensitive information from the client's private dataset $\mathcal{D}_p$ by analyzing API query patterns and generated content.

Privacy risks primarily arise from the iterative refinement process. Although the provider only observes synthetic samples and refinement queries, the sequential nature of these interactions creates attack surfaces. Prior work demonstrates that *membership inference attacks* are particularly effective in such settings [4, 8, 38], enabling adversaries to identify whether specific documents or phrases were present in the private training corpus.

We specifically address the following attack vectors in text data:

- **Membership Inference:** Determining if specific documents or rare phrases belong to $\mathcal{D}_p$.
- **Text Reconstruction:** Reconstructing sensitive text sequences from query patterns.
- **Style/Topic Inference:** Identifying writing styles, topics, or metadata characteristics.
- **Distribution Leakage:** Inferring statistical properties of the underlying text distribution.

4 Proposed Method

4.1 Overview

The DATGen framework tackles two interconnected challenges in distribution-aligned text generation: (1) identifying semantically sparse regions in embedding space without supervision, and (2) enhancing tail coverage while preserving global distributional fidelity and privacy.

As illustrated in Figure 2, our approach employs a three-stage pipeline that combines statistical analysis with privacy-preserving generation. The pipeline begins by analyzing embedding space geometry to identify underrepresented regions through statistical measures of skewness and kurtosis computed from distance distributions across clustered embeddings. This systematic approach locates tail regions without reliance on explicit supervision. Controlled Gaussian noise, constrained to sparse dimensions, then expands coverage of underrepresented semantics while preserving the geometric structure of the embedding space. Building upon these enhanced feature representations, the pipeline proceeds to generate initial synthetic samples through carefully designed prompts that incorporate domain context without privacy exposure. This initialization establishes a diverse foundation for subsequent refinement while maintaining coherence with the target distribution. The final

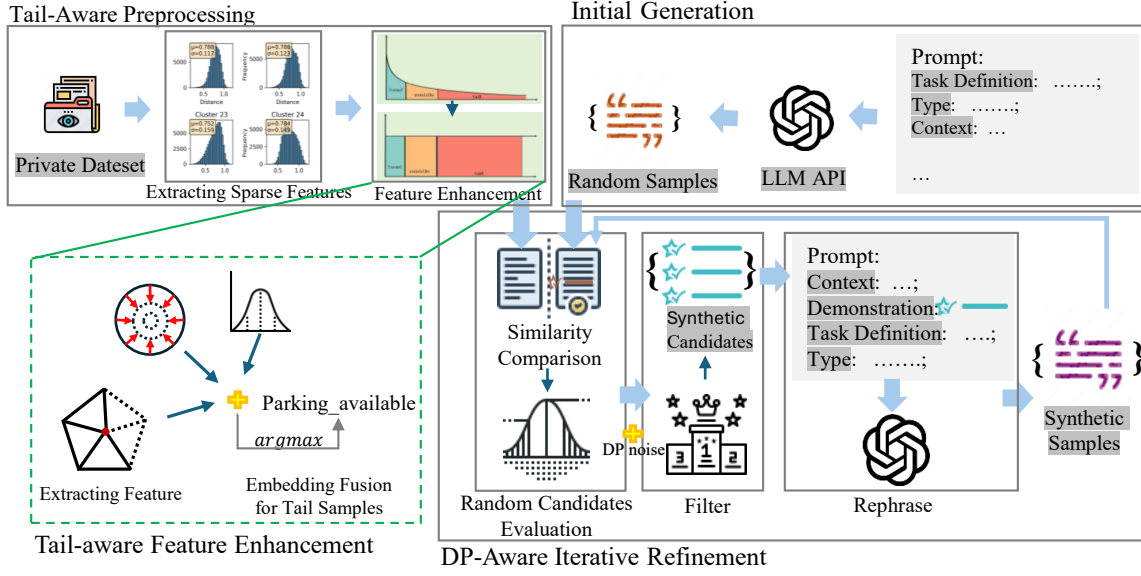


Figure 2: Overall Framework. A two-column schematic with three stages: (1) **Tail-Aware Preprocessing** detects sparse tail dimensions; (2) **DP-Aware Iterative Refinement** generates and filters synthetic samples under privacy constraints; (3) **Embedding Fusion** transfers semantics from head-dense to tail-sparse regions. Arrows indicate information flow and that tail coverage increases toward the right.

stage incorporates differential privacy guarantees during iterative improvement to balance enhancement with privacy preservation. The alignment mechanism computes cosine similarities between enhanced private embeddings and synthetic samples, applies Laplace noise to the similarity histograms, and selectively refines samples demonstrating the strongest tail feature alignment. This ensures progressive improvement in distributional fidelity within formal privacy bounds.

The coordinated pipeline enables DASGen to overcome key identification and enhancement challenges, generating synthetic text that maintains human-level semantic richness and statistical properties while providing rigorous privacy guarantees.

4.2 Tail-Aware Synthetic Data Generation

Step 1 – Tail-Aware Preprocessing with Feature Enhancement: Given a private dataset $\mathcal{D}_p = \{x_i\}_{i=1}^N$, our method first identifies sparse semantic structures through clustering in the embedding space, followed by structures feature enhancement through embedding fusion and noise augmentation to improve diversity in tail regions.

Extracting Sparse Features: For each instance x_i , we split it into sentence-level segments $\{x_{i,j}\}_{j=1}^{n_i}$ using standard sentence tokenization tools NLTK Punkt. Each segment $x_{i,j}$ is mapped to an embedding vector $\{e_{i,j}\} \in \mathbb{R}^d$ via a pretrained sentence encoder f_{RoBERTa} :

$$e_{i,j} = f_{\text{RoBERTa}}(x_{i,j}). \quad (1)$$

The set of all sentence embeddings forms a matrix $E \in \mathbb{R}^{M \times d}$, where $M = \sum_{i=1}^N n_i$ is the total number of sentence segments.

To capture the global semantic structure of the embedding space, we apply K -means clustering over E , yielding a set of cluster centroids $C = \{c_1, \dots, c_K\}$. For each embedding $e_{i,j}$, we compute its distance vector to all cluster centers:

$$d_{i,j} = [\|e_{i,j} - c_1\|, \dots, \|e_{i,j} - c_K\|]. \quad (2)$$

Then, we evaluate the skewness and kurtosis of the distance distribution $d_{i,j}$, which are indicative of irregular or underrepresented semantic patterns:

$$\text{Skewness}(d_{i,j}) = \frac{1}{K} \sum_{k=1}^K \left(\frac{d_{i,j,k} - \mu_{i,j}}{\sigma_{i,j}} \right)^3, \quad (3)$$

$$\text{Kurtosis}(d_{i,j}) = \frac{1}{K} \sum_{k=1}^K \left(\frac{d_{i,j,k} - \mu_{i,j}}{\sigma_{i,j}} \right)^4, \quad (4)$$

where $\mu_{i,j}$ and $\sigma_{i,j}$ denote the mean and standard deviation of $d_{i,j}$, respectively. We jointly leverage skewness and kurtosis to assess the asymmetry and tail heaviness of the distance distribution. In particular, a high kurtosis indicates that the embedding $e_{i,j}$ lies in a heavy-tailed region of the semantic space—exhibiting sharp peaks and outlier-like deviations—suggesting that it may correspond to rare, semantically marginal, or underrepresented content.

Embeddings satisfying $\text{Skewness}(d_{i,j}) > \tau_s$ and $\text{Kurtosis}(d_{i,j}) > \tau_k$ are marked as tail-region features. We separately maintain global tail dimensions and class-specific tail dimensions by aggregating the indices of clusters exhibiting sparse associations either globally or within specific classes.

Algorithm 1 Tail-Aware Preprocessing with Feature Enhancement

Require: Private dataset $\mathcal{D} = \{x_i\}_{i=1}^N$, Sentence encoder f_{RoBERTa} , Tail thresholds τ_s, τ_k

Ensure: Fused embeddings $\{f_{i,j}\}$ for synthetic data generation

- 1: **Sentence-level Embedding Extraction:**
- 2: **for** each $x_i \in \mathcal{D}$ **do**
- 3: Split x_i into $\{x_{i,j}\}_{j=1}^{n_i}$ via sentence tokenizer
- 4: Compute embeddings $\mathbf{e}_{i,j} = f_{\text{RoBERTa}}(x_{i,j})$
- 5: **end for**
- 6: Construct embedding matrix $\mathbf{E} \in \mathbb{R}^{M \times d}$, where $M = \sum_i n_i$
- 7: **Sparse Feature Identification via Clustering:**
- 8: Perform K -means on $\mathbf{E}$ to obtain cluster centers $\mathbf{C} = \{\mathbf{c}_k\}_{k=1}^K$
- 9: **for** each $\mathbf{e}_{i,j}$ **do**
- 10: Compute distance vector $\mathbf{d}_{i,j} = [\|\mathbf{e}_{i,j} - \mathbf{c}_k\|]_{k=1}^K$
- 11: Evaluate skewness and kurtosis of $\mathbf{d}_{i,j}$
- 12: **if** $\text{Skew}(\mathbf{d}_{i,j}) > \tau_s$ **and** $\text{Kurt}(\mathbf{d}_{i,j}) > \tau_k$ **then**
- 13: Mark $\mathbf{e}_{i,j}$ as tail embedding
- 14: **end if**
- 15: **end for**
- 16: Identify global and class-specific tail dimensions
- 17: **Global & Class-Specific Sparse Feature Aggregation:**
- 18: Compute $\bar{\mathbf{s}}_{\text{global}}$ and $\{\bar{\mathbf{s}}_{\text{class}}^{(c)}\}$ via Eqs. (5)-(6)
- 19: **Noise-Augmented Expansion:**
- 20: **for** each $\mathbf{e}_{i,j}$ marked as tail **do**
- 21: Obtain head-to-tail max distance ε_h via Eq. (7)
- 22: Sample $\gamma \sim \text{Uniform}[0, \varepsilon_h]$
- 23: Generate noise $\mathbf{s}_{i,j}^{\text{noise}} = \mathbf{e}_{i,j} + \gamma \cdot \mathcal{N}(0, \sigma^2 \mathbf{I}_{\text{tail}})$
- 24: Construct $\mathbf{s}_{i,j}^{\text{aug}} = \bar{\mathbf{s}}_{\text{global}} + \sum_c \bar{\mathbf{s}}_{\text{class}}^{(c)} + \mathbf{s}_{i,j}^{\text{noise}}$
- 25: **end for**
- 26: **Embedding Fusion:**
- 27: **for** each $\mathbf{e}_{i,j}$ **do**
- 28: Fuse via: $\mathbf{f}_{i,j} = \tanh(\alpha \mathbf{e}_{i,j} + (1 - \alpha) \mathbf{s}_{i,j}^{\text{aug}})$
- 29: **end for**
- 30: **return** Fused embeddings $\{f_{i,j}\}$

Algorithm 2 DP-Aware Synthetic Data Iterative Refinement

Require: Seed samples $\{\tilde{x}_j\}$, Private fused embeddings $\{f_{i,j}\}$

- 1: **for** each iteration $t = 1 \dots T$ **do**
- 2: **for** each synthetic sample $\tilde{x}_j$ **do**
- 3: Compute similarity scores $s_{i,j} = \cos(\mathbf{f}_i, \tilde{\mathbf{e}}_j)$
- 4: **end for**
- 5: Construct histogram $\mathbf{h} \in \mathbb{R}^B$ by discretizing $s_{i,j} \in [-1, 1]$ into B equal-width bins
- 6: Perturb histogram with Laplace noise $\epsilon \sim \text{Lap}(\lambda)$ # Ensure DP
- 7: Select top- k seeds based on noisy histogram
- 8: Generate rephrased texts via LLM API $\mathcal{R}(\cdot)$
- 9: Update embeddings for $\{\tilde{x}_j\}$ via f_{RoBERTa}
- 10: **end for**
- 11: **return** Final synthetic dataset $\tilde{\mathcal{D}}$

Tail-Aware Feature Enhancement: To address the deficiency of long-tail representations, we enhance sparse features in the embedding space by integrating global and class-specific information with controlled noise perturbation.

(i) Injection of global and class-specific sparse feature: For each tail region, we compute the global mean sparse embedding $\bar{\mathbf{s}}_{\text{global}}$ and class-specific mean sparse embedding $\bar{\mathbf{s}}_{\text{class}}^{(c)}$ over the identified sparse dimensions:

$$\bar{\mathbf{s}}_{\text{global}} = \frac{1}{|T_{\text{global}}|} \sum_{(i,j) \in T_{\text{global}}} \mathbf{e}_{i,j}, \quad (5)$$

$$\bar{\mathbf{s}}_{\text{class}}^{(c)} = \frac{1}{|T_{\text{class}}^{(c)}|} \sum_{(i,j) \in T_{\text{class}}^{(c)}} \mathbf{e}_{i,j}. \quad (6)$$

These aggregated embeddings capture the semantic properties of underrepresented regions and provide auxiliary signals to enrich the embedding space.

(ii) **Noise-Augmented Embedding Expansion:** Recognizing the inherent complexity of natural language, we introduce a controlled noise mechanism to enhance diversity in the embedding space. Rather than applying isotropic Gaussian noise, which may distort the semantic structure, we design an anisotropic perturbation strategy that leverages the observed geometric spread of head clusters [35]. Specifically, for each head cluster center $\mathbf{c}_h$, we measure its maximal distance to the farthest in-cluster embedding $\mathbf{e}'_h$ to estimate the semantic dispersion of head regions:

$$\varepsilon_h = \max_{\mathbf{e}'_h \in C_{\text{head}}} \|\mathbf{c}_h - \mathbf{e}'_h\|. \quad (7)$$

We then assume that tail clusters should exhibit a similar degree of dispersion, i.e., $\varepsilon_t \approx \varepsilon_h$, where ε_t denotes the estimated dispersion in tail regions. Such an assumption allows us to transfer the structural variance of head clusters into the tail space, enriching the representation diversity without explicit supervision.

For each tail embedding $\mathbf{e}_{i,j}$, we sample a noise magnitude $\gamma \sim \text{Uniform}[0, \varepsilon_h]$, and apply directionally consistent noise restricted to tail-sensitive dimensions:

$$\mathbf{s}_{i,j}^{\text{noise}} = \mathbf{e}_{i,j} + \gamma \cdot \mathcal{N}(0, \sigma^2 \mathbf{I}_{\text{tail}}), \quad (8)$$

where $\mathcal{N}(0, \sigma^2 \mathbf{I}_{\text{tail}})$ denotes a d -dimensional Gaussian noise with zero mean and covariance matrix $\sigma^2 \mathbf{I}_{\text{tail}}$. The projection matrix $\mathbf{I}_{\text{tail}} \in \mathbb{R}^{d \times d}$ is diagonal, with 1s at tail-sensitive dimensions and 0s elsewhere, ensuring perturbation is only applied to sparse features in the semantic space.

Remark: This head-to-tail variance transfer provides a statistical prior for shaping the distribution of synthetic tail features, enabling better generalization while preserving semantic plausibility.

(iii) **Embedding fusion:** We integrate the original embeddings $\mathbf{e}_{i,j}$ with sparse-enhanced representations $\mathbf{s}_{i,j}^{\text{aug}}$ using a nonlinear fusion strategy:

$$\mathbf{f}_{i,j} = \phi(\alpha \mathbf{e}_{i,j} + (1 - \alpha) \mathbf{s}_{i,j}^{\text{aug}}), \quad (9)$$

where ϕ denotes an activation function controlling the smoothness and nonlinearity of the fused embedding. Here, $\mathbf{s}_{i,j}^{\text{aug}} = \bar{\mathbf{s}}_{\text{global}} + \sum_c \bar{\mathbf{s}}_{\text{class}}^{(c)} + \mathbf{s}_{i,j}^{\text{noise}}$ represents the final sparse-enhanced representation combining global structure, class-specific semantics, and controlled noise.

Empirically, we adopt tanh as the default fusion strategy due to its effectiveness in balancing distributional smoothness (e.g., lower skewness and kurtosis) and representation quality. The fused embeddings $\mathbf{f}_{i,j}$ serve as refined semantic anchors to guide subsequent synthetic data generation and selection, ensuring that the generated samples preserve both common and rare patterns in the representation space.

Step 2 – Initial Synthetic Data Generation: To bootstrap synthetic examples via prompting or public corpora, we initialize synthetic seed samples $\{\tilde{x}_j\}_{j=1}^M$ from a public domain or via manually crafted prompts. These samples serve as the starting point for downstream refinement. Embeddings $\tilde{\mathbf{e}}_j$ are obtained using the same encoder f_{RoBERTa} .

Step 3 – DP-aware Iterative Refinement: In the final stage, we refine synthetic seeds under DP constraints. Specifically, we adopt the Laplace mechanism to ensure ϵ -DP during the alignment phase. For each iteration, we compute cosine similarities between private embeddings $\{\mathbf{f}_i\}$ and synthetic seeds $\{\tilde{\mathbf{e}}_j\}$:

$$s_{ij} = \cos(\mathbf{f}_i, \tilde{\mathbf{e}}_j). \quad (10)$$

These similarity scores s_{ij} are then discretized into B bins to form a histogram $\mathbf{h} \in \mathbb{R}^B$: $h_b = |\{(i, j) \mid s_{ij} \in \text{Bin}_b\}|$, for $b = \{1, \dots, B\}$. To preserve privacy, we apply the Laplace mechanism by adding independently sampled noise $\epsilon_b \sim \text{Lap}(\Delta/\epsilon)$ to each bin count:

$$\hat{h}_b = h_b + \epsilon_b. \quad (11)$$

where Δ denotes the global sensitivity of the histogram query. The privacy budget ϵ controls the noise scale and is distributed evenly across all bins. If refinement is performed for T iterations, the total budget ϵ_{total} is divided such that each iteration consumes $\epsilon = \epsilon_{\text{total}}/T$. From the privatized histogram $\hat{\mathbf{h}}$, we select the top- k bins with highest noisy counts, corresponding to the strongest estimated alignments between private and synthetic embeddings. Synthetic seeds are then updated accordingly using these high-confidence pairs in each iteration.

Next, for the selected synthetic seeds $\tilde{x}_j$ from these top- k bins, we perform iterative rephrasing via LLM prompting to improve diversity. These newly generated texts $\tilde{x}_j^{(t+1)}$ are re-encoded to obtain updated embeddings $\tilde{\mathbf{e}}_j^{(t+1)}$: $\tilde{\mathbf{e}}_j^{(t+1)} = f_{\text{RoBERTa}}(\tilde{x}_j^{(t+1)})$. The similarity scoring and DP sampling process is then repeated for the next iteration $t + 1$.

This refinement loop continues until reaching the predefined number of iterations. By applying sequential composition 3.3, DAS-GEN guarantees that the synthetic data remains differentially private with respect to the original sensitive data.

5 Evaluation and Discussion

5.1 Experiment Settings

All experiments utilize the HuggingFace Transformers library, where GPT-2 is fine-tuned using the AdamW optimizer with a learning rate of 2×10^{-5} , a batch size of 32, and 5 training epochs applied uniformly across all tasks.

All DP-related methods utilize DP-SGD where applicable, with privacy budgets set to $\epsilon = 1$, following [46]. Experiments are conducted on an NVIDIA 4090 GPU. All reported data are based on the average results of three experiments for stability.

Comparison Methods. We compare DAS-GEN against two representative baselines:

- **Fine-tuning-based Method.** A synthetic data generation method [49], where the synthetic data is used to further fine-tune the data generator itself. Prior work (hereafter referred to as Fine-tune) demonstrates that synthetic data can negatively impact model performance due to insufficient representation of long-tail distributions [39]. This work is included as a baseline to underscore the impact of the long-tail issue in synthetic data generation.
- **Prompt-based Method.** A synthetic data generation method that produces synthetic data via API access to foundation models without model fine-tuning. AugPE [46] demonstrates that prompt-based methods are effective in improving the quality of synthetic data. By comparing DAS-GEN with AugPE, we demonstrate that explicitly addressing the long-tail distribution leads to improved synthetic data quality and provides better resilience against model collapse.

Evaluation Metrics. We use the following metrics:

- **Tail Coverage:** Let $\mathcal{T} = \{\mathbf{z}_1, \dots, \mathbf{z}_m\}$ be the set of tail embeddings from the real data (defined as the bottom 10% by density via KNN density estimation) and $\mathcal{S} = \{\tilde{\mathbf{z}}_1, \dots, \tilde{\mathbf{z}}_n\}$ be the embeddings of synthetic data. The tail coverage is:

$$C_{\text{tail}} = \frac{|\{\mathbf{z} \in \mathcal{T} \mid \exists \tilde{\mathbf{z}} \in \mathcal{S}, \cos(\mathbf{z}, \tilde{\mathbf{z}}) > 0.3\}|}{|\mathcal{T}|}$$

This metric measures the proportion of the tail region in the real data that is covered by the synthetic data.

- **Perplexity (PPL)** [39]: Perplexity quantifies a language model's uncertainty when evaluating a text sequence. For a token sequence $\mathbf{W} = (w_1, w_2, \dots, w_N)$, PPL is defined as:

$$\text{PPL}(\mathbf{W}) = \exp \left(-\frac{1}{N} \sum_{i=1}^N \log P_{\mathcal{M}}(w_i | w_1, \dots, w_{i-1}) \right)$$

where $P_{\mathcal{M}}(w_i | w_1, \dots, w_{i-1})$ is the conditional probability of token w_i given all previous tokens, as estimated by the evaluator model $\mathcal{M}$. We use a fixed GPT-2 model¹ as a consistent evaluator across all synthetic datasets. Lower PPL indicates higher fluency, while higher PPL may reflect diversity or incoherence. The distribution of PPL scores across synthetic datasets reveals distributional shifts; a severely left-shifted distribution indicates distribution collapse, while a distribution matching human text suggests balanced diversity.

Models. We use RoBERTa [28] text embeddings. All methods use the same GPT-2-based generators (GPT2, GPT2-Medium and GPT2-large) [32] for fair comparison.

Datasets. We evaluate our approach on two real-world datasets, both exhibiting typical long-tailed distributions [35]. (i) Yelp Dataset [20]: Following prior work on DP synthetic data [46], we adopt the Yelp dataset containing 1.9M training samples, with 5K samples each for

¹<https://huggingface.co/openai-community/gpt2>

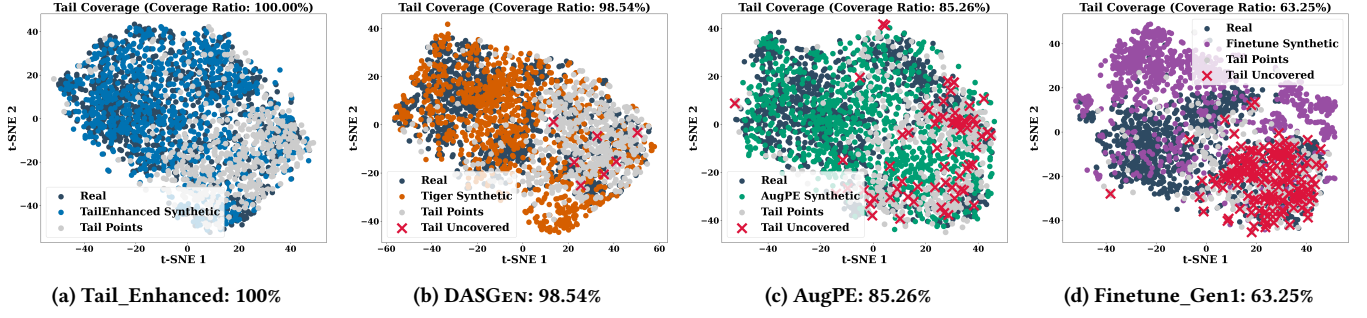
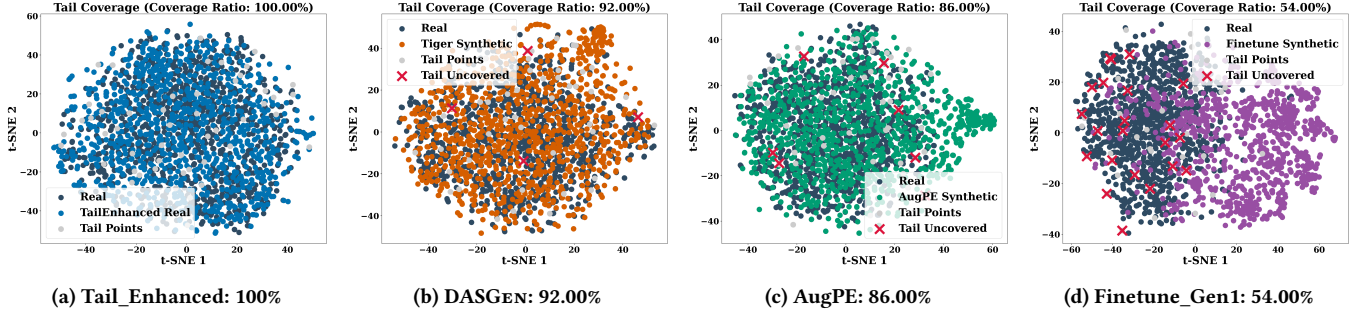
Figure 3: Tail coverage on the Yelp dataset. Red $\times$ marks tail points not covered by any synthetic sample.

Figure 4: Tail coverage on the ICLR'25 dataset.

validation and testing. (ii) ICLR'25 Dataset: We further construct a dataset by collecting recent peer reviews of ICLR'2025 submissions from OpenReview², reflecting data published post recent LLM releases. The ICLR'25 dataset comprises 28K training instances, with 9K validation and test examples each.

Hyperparameter Selection. In our evaluation, tail regions are defined as the bottom 10% of training examples ranked by their embedding-space density, estimated via KNN density estimation. More details, we choose the number of K-means clusters K based on the trade-off between semantic granularity and the statistical stability of cluster-level features. For large corpora (e.g., Yelp, 1.9M sentences), we set $K = 500$ to achieve fine-grained partitioning of the embedding space. This value also ensures that each cluster retains enough samples for reliable skewness and kurtosis estimation. For smaller datasets (e.g., ICLR'25, 28k samples), we set $K = 100$, which provides a balanced trade-off between computational efficiency and representational granularity. The cosine similarity threshold for tail coverage is set to 0.3, following the common practice in semantic textual similarity [23].

5.2 Main Results

5.2.1 Evaluation of Long-tail Fidelity. Figures 3 and 4 present t-SNE visualizations for one feature-enhanced space (i.e., Tail_Enhanced) and three synthetic data generation strategies: DASGEN, AugPE, and Finetune. Each plot displays real samples (grey point), and synthetic samples (colorful point). Uncovered tails—those with no synthetic sample within the threshold—are marked with a red cross

“ $\times$ ”. Among these methods, *Tail_Enhanced* achieves full coverage, while DASGEN reaches 98.54%. In contrast, AugPE and Finetune cover only 85.26% and 63.25% of tail points, respectively, leaving many regions underrepresented. Notably, the generated samples are not merely clustered in high-density regions but are distributed toward the periphery of embedding space. This suggests improved coverage and greater semantic diversity in the synthetic data.

As shown in Table 1, we regard human-written PPL as a reference anchor: synthetic samples with comparable PPL are considered linguistically plausible, while excessively low PPL may indicate collapsed diversity, and excessively high PPL implies degraded fluency. Across both datasets, DASGEN consistently demonstrates closer perplexity to compared to baseline fine-tuned GPT2 models and the AugPE-enhanced generator. On the Yelp dataset, for example, GPT2-Large+DASGEN achieves a median PPL of 44.18. This indicates that DASGEN produces samples with more balanced semantic distribution and greater diversity. A similar result is observed on the ICLR'25 dataset, where DASGEN maintains favorable trends in preliminary evaluations, suggesting that it effectively smooths the representation distribution and suppresses extreme outliers in the embedding space. These results highlight the advantage of our tail-aware augmentation strategy in generating high-quality and robust synthetic data.

Evaluation of Downstream Tasks We assess whether synthetic samples can enhance model performance on downstream tasks.

(i) *Classification Accuracy.* We evaluate the utility of synthetic data by fine-tuning GPT2 on synthetic datasets and testing performance on real test sets. Two benchmark tasks are considered for the Yelp dataset: {Business Category} and {Review Rating}, and two tasks for

²<https://openreview.net/group?id=ICLR.cc/2025/Conference>

Table 1: Perplexity of Generated Data Points.

Dataset: Yelp			
Generators	Methods	Median PPL	Δ PPL w.r.t. Human
Human-written		43.31	–
GPT2_Gen1	Fine-tune	50.95	+ 7.64
GPT2_Gen2	Fine-tune	21.41	- 21.90
GPT2	AugPE	35.34	- 7.97
GPT2	DASGEN	38.77	- 4.54
GPT2-Medium	AugPE	45.37	+ 2.06
GPT2-Medium	DASGEN	44.59	+ 1.28
GPT2-Large	AugPE	56.85	+ 13.54
GPT2-Large	DASGEN	44.18	+ 0.87

Dataset: ICLR'25			
Human-written		62.94	–
GPT2_Gen1	Fine-tune	23.58	-39.36
GPT2_Gen2	Fine-tune	17.65	+ 45.29
GPT2	AugPE	25.97	- 36.97
GPT2	DASGEN	38.19	-24.75
GPT2-Medium	AugPE	37.87	- 25.07
GPT2-Medium	DASGEN	48.38	-14.56
GPT2-Large	AugPE	275.76	+ 212.82
GPT2-Large	DASGEN	68.56	+5.62

the ICLR'25 dataset: {Research Area} and {Recommendation Score}. We compare DASGEN against both Fine-tune and AugPE under two DP budgets: $\epsilon = 4$ and $\epsilon = 1$.

Table 2: Comparison of Downstream Model Accuracy under DP budgets $\epsilon = 4$ and $\epsilon = 1$.

Dataset: Yelp					
Generators	Methods	$\epsilon = 4$		$\epsilon = 1$	
		Rating	Category	Rating	Category
GPT2_Gen1	Fine-tune	67.5	72.8	66.8	71.8
GPT2_Gen2	Fine-tune	64.8	68.1	61.8	64.4
GPT2	AugPE	66.4	74.9	66.9	74.4
GPT2	DASGEN	68.6	75.1	68.7	74.7
GPT2-Medium	AugPE	66.8	74.9	66.9	74.4
GPT2-Medium	DASGEN	68.9	75.1	68.6	74.6
GPT2-Large	AugPE	67.3	74.4	66.5	75.0
GPT2-Large	DASGEN	69.8	75.1	68.7	75.1

Dataset: ICLR'25					
Generators	Methods	$\epsilon = 4$		$\epsilon = 1$	
		Area	Rating	Area	Rating
GPT2_Gen1	Fine-tune	32.1	32.3	30.5	32.1
GPT2_Gen2	Fine-tune	31.9	32.0	30.3	32.0
GPT2	AugPE	39.9	37.6	36.7	34.5
GPT2	DASGEN	41.8	39.5	38.7	36.0
GPT2-Medium	AugPE	40.3	38.3	37.1	34.9
GPT2-Medium	DASGEN	42.5	39.0	38.5	35.5
GPT2-Large	AugPE	41.5	38.7	38.0	35.2
GPT2-Large	DASGEN	43.5	40.5	39.5	36.0

As shown in Table 2, DASGEN outperforms all baselines across both datasets and privacy settings. For example, on the Yelp dataset under $\epsilon = 1$, DASGEN improves the rating classification accuracy from 66.8% to 68.7%, and the category classification accuracy from 71.8% to 74.7%. Similarly, on the ICLR'25 dataset, DASGEN achieves the best results with 39.5% on rating prediction and 43.5% for area classification under $\epsilon = 4$. Notably, across all model sizes, DASGEN surpasses Fine-tune and AugPE, indicating that DASGEN better supports downstream learning under strict privacy constraints.

Table 3: PPL Improvement over AugPE of GPT2-Series Models on Different Synthetic Data.

Models	over AugPE		
	Gen_1	Gen_2	Gen_3
GPT2	2.55×	7.28×	15.96×
GPT2-Medium	2.35×	6.94×	14.50×
GPT2-Large	2.12×	5.58×	14.00×

(ii) *Easing Model Collapse*. Model collapse is a phenomenon in which iterative training on synthetic data amplifies frequent patterns while progressively diminishing the model's ability to represent diverse linguistic features. Perplexity is a widely adopted indicator of model collapse, with higher perplexity suggesting a more severe collapse. Table 3 presents the perplexity and AugPE on the Yelp dataset as multiples relative to DASGEN, where each Gen_i denotes a model trained on synthetic data generated at the i -th iteration. All models were trained using the hyperparameter settings from [52].

As shown in Table 3, DASGEN demonstrates superior effectiveness in mitigating *model collapse* compared to Fine-tune and AugPE. For instance, on Gen_3 , the perplexity of AugPE is 15.96×, 14.50×, and 14.00× higher than that of DASGEN for GPT2, GPT2-Medium, and GPT2-Large, respectively. These results indicate that DASGEN significantly outperforms AugPE in preventing model collapse, particularly under deeper generation iterations. These findings highlight the critical need for control mechanisms when utilizing synthetic data in iterative training pipelines to prevent quality degradation and model collapse.

6 Conclusion and Future Work

This paper tackles the challenge of *distributional alignment* in synthetic text generation, with a focus on preserving long-tail characteristics of real-world data. We propose DASGEN, a tail-aware framework that enhances rare-feature coverage in the embedding space via prompt-based steering and privacy-preserving refinement, without requiring model fine-tuning. Extensive experiments show that DASGEN consistently improves tail coverage while maintaining strong downstream performance, demonstrating its robustness and practical value.

Future work will extend this study in two directions. We plan to evaluate DASGEN across a wider range of state-of-the-art large language models to further assess its generalizability. In addition, we will explore automated hyperparameter optimization strategies, such as adaptive selection of cluster count K and tail threshold τ , to improve usability and adaptability in diverse application scenarios.

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A Additional Datasets Details

For the ICLR’25 dataset, we collected the metadata of each review via the OpenReview Python API³. The textual fields *summary_of_the_paper*, *strength_and_weaknesses*, and *summary_of_the_review* were concatenated in sequence to form a single review text. Two categorical attributes—the research area and the recommendation score—were used as labels. After this preprocessing step, the per-class sample counts are shown in Tables 4 and 5.

Table 4: Research Area Label Statistics of ICLR’25 Dataset.

Research Area	Count	Proportion
applications	10,727	22.95%
generative models	4278	9.15%
fairness and privacy	4225	9.04%
unsupervised and self-supervised learning	3458	7.40%
datasets and benchmarks	3051	6.52%
reinforcement learning	2661	5.69%
general machine learning	2229	4.77%
applications to physical sciences	2204	4.71%
explainable AI	2059	4.40%
optimization	2029	4.34%
learning on graphs	1759	3.76%
transfer learning	1493	3.19%
learning theory	1460	3.12%
learning on time series and dynamical systems	1085	2.32%
applications to robotics	973	2.08%
probabilistic methods	961	2.06%
applications to neuroscience	855	1.83%
causal reasoning	471	1.01%
hybrid AI systems	425	0.91%
hardware	345	0.74%

B Prompts in Experiments

In both the initial generation phase and the iterative refinement phase, we adopt a consistent prompt design for the GPT2-series models, with task-specific adaptations for two application domains: Yelp business review generation and ICLR’25 academic review generation.

Initial Generation Phase: In the Yelp scenario, the model is instructed to generate an example review based on the provided {Business Category} and {Review Rating}, where the rating falls within the range of 1.0 to 5.0. The input follows the format:

“Business Category: {label_1} || Review Rating: {label_2} with keyword {subcategory}”

where subcategory provides an additional keyword or subcategory. In the ICLR’25 scenario, the model is instructed to generate an academic review given the {Research Area} and the {Recommendation Rating} of a paper. The input follows the format:

“Suppose that you are a {reviewer}. Write a paper review based on Research Area: {label_1} || Recommendation: {label_2}”

Iterative Refinement Phase: In this phase, the model is guided to rephrase an existing review in a specified tone, conditioned on task-specific metadata. For the Yelp dataset, the prompt template is:

³<https://github.com/openreview/openreview-py>

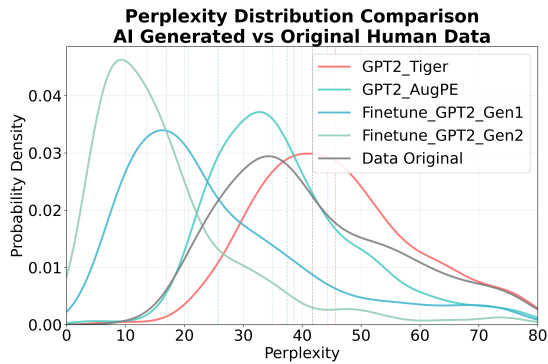


Figure 5: PPL of Generated Data Points

“Based on “Business Category: {label_1} || Review Stars: {label_2}”, please rephrase the following sentences {in a selected_tone}: {input}”

For the OpenReview dataset, the prompt template is:

“Based on “Area: {label_1} || Recommendation: {label_2}”, please rephrase the following sentences {in a selected_tone}: {input}”

C PPL of Generated Data Points

Table 5: Recommendation Score Label Statistics of ICLR’25 Dataset.

Recommendation Score	Count	Proportion
Recommendation: 6: marginally above the acceptance threshold	14695	31.05%
Recommendation: 5: marginally below the acceptance threshold	13101	27.68%
Recommendation: 3: reject, not good enough	11537	24.37%
Recommendation: 8: accept, good paper	6214	13.12%
Recommendation: 1: strong reject	1029	2.17%
Recommendation: 10: strong accept	172	0.36%

Figure 5 serves as a visual complement to Table 1 in Section 5.2, presenting the probability density distributions of PPL for synthetic datasets produced by different strategies, alongside the original human-written dataset.

The human-written dataset exhibits a moderately concentrated distribution, reflecting natural linguistic variability. In contrast, GPT2_TIGER displays a right-shifted distribution with broader coverage across the high-perplexity region, indicating effective supplementation of sparse (tail) features while maintaining overall fluency. Fine-tuned GPT2_GEN1 and GPT2_GEN2 show markedly lower perplexity peaks, implying overfitting and reduced variability, consistent with early-stage model collapse. Notably, GPT2_GEN2 exhibits the narrowest and most left-skewed distribution, reflecting severe diversity degradation under iterative generation. These observations confirm that iterative fine-tuning on synthetic data narrows and left-shifts the perplexity distribution, diminishing distributional diversity and tail coverage. In contrast, TIGER preserve human-like perplexity levels while significantly enhancing the representation of sparse features, thereby mitigating model collapse in long-term synthetic data generation.